32-bit microcontrollers

DAC Module

Applicable to:

Series	Product Model	Series	Product Model
HC32L17	HC32L176PATA HC32L176MATA HC32L176KATA HC32L176JATA HC32L170JATA HC32L170FAUA	**HC32F17**	HC32F176PATA HC32F176MATA HC32F176KATA HC32F176JATA HC32F170JATA HC32F170FAUA
HC32L19	HC32L196PCTA HC32L196MCTA HC32L196KCTA HC32L196JCTA HC32L190JCTA HC32L190FCUA	**HC32F19**	HC32F196PCTA HC32F196MCTA HC32F196KCTA HC32F196JCTA HC32F190JCTA HC32F190FCUA

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1. Abstract

This application note primarily introduces the use of the DAC module in Huada Semiconductor's MCU*.

Note:

- This application note is supplementary material for Huada Semiconductor's MCU* and cannot replace the user manual. For specific functions and register operations, please refer to the user manual.

2. Function Introduction

This note will explain the application and usage precautions of Huada Semiconductor's MCU* DAC module. This module integrates a 12-bit DAC converter with one output channel, DAC_OUT. This output can be connected to an external port, used as the positive input signal of an OPA, or as the analog input of an ADC. The DAC can provide four reference sources: AVCC voltage, ExRef pin, built-in 1.5V reference voltage, and built-in 2.5V reference voltage.

Supported models are listed on the cover.

3. DAC Module

3.1 DAC Voltage Output

The DAC module converts the digital input to an output voltage between 0 and VREF+.

The analog output voltage of the DAC channel is determined by the following formula:

$$DAC_{out} = V_{ref} \quad * \quad DOR/4096$$

3.2 DAC Trigger Selection

If the TEN0 control bit is set to 1, the conversion can be triggered by an external event or software. The TSEL[2:0] control bits determine the selected trigger conversion. Refer to Figure 1 for the trigger source. When the DAC selects the timer TRGO output or the rising edge of an external port, the DAC module updates the DAC register after three APB1 cycles.

If software triggering occurs, the conversion begins.

TSEL	000	001	010	011	100	101	110	111
trigger source	TIM0_ TRADC	TIM1_ TRADC	TIM2_ TRADC	TIM3_ TRADC	TIM4_ TRADC	TIM5_ TRADC	SW TRIG	EXTI

Figure 1 Trigger Source Graph

3.3 DMA Request

When DAMEN0 is set to 1, a DAC DMA request is generated if a transmit trigger occurs. The value of the DAC_DHR0 register is passed to the DAC_DOR0 register.

Since the DAC DMA request has no buffer queue, a DAM channel underflow may occur. The corresponding flag DAMUDR0 is set to 1, DAM data transmission stops, but the DAC channel continues converting existing data. Software writes a "1" to clear the DMAUDR0 flag, clears the DMAEN bit of the used DMA data stream, and reinitializes the DMA and DAC channels to begin correct DMA transmission again.

3.4 DAC Triangle Wave Generation

A small-amplitude triangular wave is superimposed on the DC signal. WAVE0[1:0] is set to "10", and the amplitude is configured via the MAP0[3:0] bits of the DAC_CR register. After each trigger event, the internal triangular wave counter increments after three APB clock cycles. Once the configured amplitude is reached, the counter decrements to zero and then increments again.

3.5 DAC Noise Generation

To generate pseudo-noise with variable amplitude, LFSR can be used, setting WAVE0[1:0] to "01". The preload value in LFSR is 0xAAA. After each trigger event, this register is updated according to a specific algorithm after three APB clock cycles.

4. Reference Examples and Drivers

Through the above introduction, and with the corresponding user manuals, we have gained a further understanding of the functions and operation methods of the DAC module in the Huada Semiconductor MCU*.

Huada Semiconductor (HDSC) also provides a driver library for this module. Users can further familiarize themselves with the module and the application of the driver library by opening the sample project. In actual development, users can also directly refer to the sample and use the driver library to quickly implement the operation of this module.

5. Summary

The above chapters briefly introduce the DAC module of the Huada Semiconductor MCU* and explain the use of the DAC module in detail. In actual application development, if users need to further understand the usage methods and operation of this module, they should refer to the corresponding user manual. The driver library mentioned in this article can be used for further experimentation and learning, or it can be directly applied in actual development.

6. Other Information

Technical support information: www.hdsc.com.cn

Supported models are listed on the cover.

7. Version Information & Contact Us

Date	Version	Change History
2019/6/24	Rev1.0	Initial release.

If you have any comments or suggestions during the purchase and use process, please feel free to contact us.

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